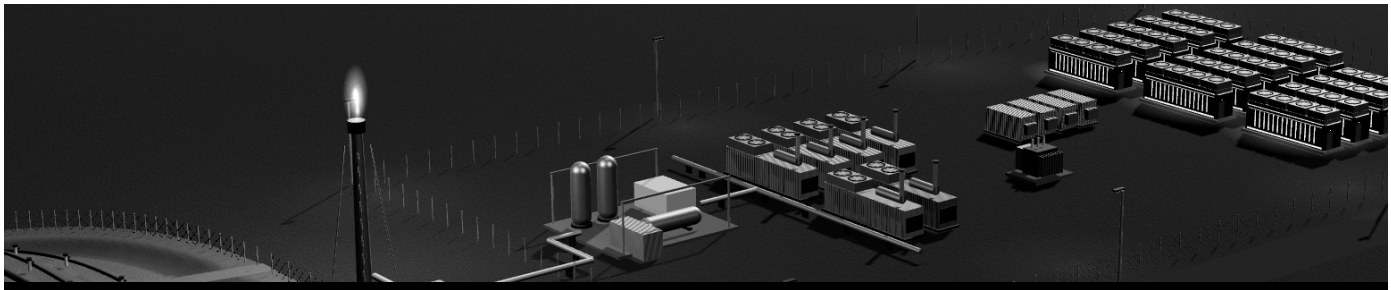
Monarch Hill Landfill, Pompano Beach, Florida: a first sizing

What the gas the site now flares could power, how much land it would take, and what it could earn each year.

15.3 MW	14	23 ac	1,397k
POWER FROM THE GAS	COMPUTE MODULES	LAND NEEDED	MMBTU BOUGHT / YR

GAS SALE, PER YEAR	LAND LEASE, PER YEAR	TOTAL TO THE OWNER, PER YEAR
up to $1.05M	$23k to $47k	$23k to $1.09M



YOUR SITE IN OUR LIVE 3D MODEL — ACCRETIONCOMPUTE.COM/SITE-MODEL?SRC=LFG&FLOW=6912&CH4=39.6

THE MATH

Landfill gas flared beside the turbine plant (2022)	6,912 scfm
Methane content (2023)	39.6%
Energy in the gas	167.9 MMBtu per hour
Power at a heat rate of 11,000 Btu/kWh	about 15.3 MW
Compute (power less cooling and losses, /1.2)	about 12.7 MW
Compute modules (about 1 MW each)	14
Gas engines (running plus one spare)	6 at 3.3 MW
Land: 5 acres plus 1.2 acres per MW	about 23 acres
Gas bought per year (95% run time)	1,397k MMBtu
Gas payment, up to $0.75/MMBtu	up to $1.05M
Land lease at $1,000 to $2,000 per acre per year	$23k to $47k

Landfill gas: about 1,012 Btu per cubic foot of methane. Prices in US dollars; the price band is the one shown on our Site model page.

WHAT IT WOULD LOOK LIKE

- Gas engines on a small parcel near the gas, a battery to smooth the load, and factory-built compute modules leased to an AI company.
- No grid connection and no export: all of the power is used on site.
- We build and run the site. The owner puts in no capital.
- Multi-year gas and land agreements, matched to the tenant's lease, at a fixed price.

WORTH CHECKING

- Gas rights and any existing gas agreement
- Florida DEP air construction permit (Chapter 62-210)
- Tie-in under the landfill's Title V permit (NSPS Subpart XXX, NESHAP Subpart AAAA)
- County or municipal zoning for the data hall

Source: EPA Greenhouse Gas Reporting Program, 2023: about 13,125 scfm collected at 39.6% methane. EPA Landfill Methane Outreach Program: about 6,912 scfm flared in 2022 beside the turbine plant (about 4.6 MW average in 2025, EIA). This is a first read from public data, not an offer: no cost and no obligation. The real price and terms are set in the gas and land agreements after a site review.